

# PAPER\_721: UQFF Quantum Variable Documents: r\_j, d\_g, F\_U, f\_feedback, Omega\_g

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## Abstract

Five quantum variable document pages from the UQFF Knowledge Base define the foundational scalar quantities used across all UQFF master equations:  $r_j$  (magnetic string distance),  $d_g$  (Galactic Center distance),  $F_U$  (unified field strength total),  $f_{feedback}$  (AGN feedback factor), and  $\Omega_g$  (Galactic spin rate). Together they define the  $U_m$  and  $U_{bi}$  coupling to the Milky Way SMBH gravitational system.

## 1. Variable Registry

| Variable       | Value                                          | Equation Role                                                                               |
|----------------|------------------------------------------------|---------------------------------------------------------------------------------------------|
| $r_j$          | $1.496 \times 10^{13} \text{ m (1 AU)}$        | $U_m = \sum_j \frac{\mu_j}{r_j}(\dots)$                                                     |
| $d_g$          | $2.55 \times 10^{20} \text{ m (\$sim\$8 kpc)}$ | $U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} \dots$                                |
| $F_U$          | $2.28 \times 10^{65} \text{ N}$                | $F_U = \sum [k_i U_{gi} - \beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} E_{react}] + \dots$    |
| $f_{feedback}$ | 0.1                                            | $U_{g4} = k_4 \frac{\rho_{SCm} M_{bh}}{d_g} e^{-\alpha t} \cos(\pi t_n) (1 + f_{feedback})$ |
| $\Omega_g$     | $7.3 \times 10^{-16} \text{ rad/s}$            | $U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} \dots$                                |

## 2. Magnetic String $U_m$

$$U_m = \sum_j \left[ \frac{\mu_j}{r_j} (1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j \right] \cdot P_{SCm} \cdot E_{react} \cdot (1 + 10^{13} f_H) \cdot (1 + f_{quasi})$$

## 3. Buoyancy-Gravity Coupling

$$U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} (1 + \varepsilon_{sw} \rho_{sw}) U_{UA} \cos(\pi t_n)$$

## References

- UQFF KB grok\_share\_ba508f76c8e.txt entry #70
- Session 176, v5.33

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